

Annotations (74)

- 1 Highlight | Yellow (1)
- 4 Underlines | Default (4)
- 33 Outlines | Default (33)
- 36 Notes

1. What the-valley is trying to make possible

Page 26 | Note

I think it's important to embrace the symmetry of new agent understanding and new human understanding. I love that well maintained contextual information about a project is as useful for humans as agents. A project is defined by all the data that helps it meet it's current AND future goals.

Sep 21, 2026

2. Existing systems and the alternatives they make credible

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The implication for agents is to investigate coordination before the integration

Sep 21, 2026

Note:

I'm very interested in this.
I think deeply understanding modes
of collaborative project (software) editing
is in the cards. Currently it's a
race to get your change in before
the others. Because it's a branching
tree model. What are other approaches?

Sep 21, 2026

Page 40 | Highlight (Yellow)

Approval authorizes an action. Learning updates project understanding. Attention directs a limited human resource. A single review interaction can serve all three, but evaluating only approval latency would hide losses in the other two.

Sep 22, 2026

Outline (Default)

Approval authorizes an action. Learning updates project understanding. Attention directs a limited human resource. A single review interaction can serve all three, but evaluating only approval latency would hide losses in the other two.

Sep 22, 2026

3. Verification that can travel with a change

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and when its result still applies. This chapter's research conclusion is favorable to portable evidence and selective reuse. It is considerably less favorable to treating signed local execution plus eventual auditing as a substitute for a protected verification boundary in adversarial settings. Those are separate architectural choices. The first does not require the second.

Sep 22, 2026

Note:

I agree it doesn't mitigate an adversarial attack. In the majority of scenarios I don't see an adv. incentive. Any tampering would be quickly discovered after integration.

Sep 22, 2026

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it?: Whether the signer told the truth

Sep 22, 2026

Note:

The signature is for non-repudiation of the claim. Of course it doesn't guarantee the truth of the claim.

Sep 22, 2026

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The current implementation is a useful base because it already exposes the subject, predicate, signer, and

transfer decision separately. The research priority is to make each obligation explicit at those existing boundaries, before adding a reputation system whose score might conceal which obligation remains unsatisfied.

Sep 22, 2026

Note:

I think the worst set of inputs into verification policy is a strawman of a set of signals we can use for an approximating integration controller that prioritizes throughput over perfect accuracy. Especially because there is a gap between ~~computable~~ checks and human requirement so things are already imperfect!

Sep 22, 2026

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Bazel's documentation gives

Sep 22, 2026

Note:

I've pondered if Bazel is a better builder choice for the valley. For now it's package set seems too incomplete. I would be curious to use a ~~functional~~ functioning valley instance to test and bootstrap a form of Bazel 0.5

Sep 22, 2026

This is a protocol-level counterexample, not a claim that the existing operator is malicious. The current trust model can be entirely appropriate for an operator's own devices and agents. The inaccurate part is claiming that these mechanisms force a malicious signer to leave detectable evidence of how it lied.

Sep 22, 2026

Note:

This looks like an agent documentation error. It wasn't my intent in the design.

Sep 22, 2026

Recommendation: keep three deployment profiles conceptually distinct: trusted personal attestors; delegated builders whose signing authority is protected from submitted code; and independent witnesses that check selected claims. A project may combine them. A numeric score should never make the first profile appear cryptographically equivalent to the second.

Sep 22, 2026

Note:

I like this.

Sep 22, 2026

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maintenance cost. V18 — DSSE protocol Recommendation: preserve a precise internal claim model and implement an explicit interoperable representation at the boundary. Re-encoding a statement changes its signed bytes. An adapter should either preserve the original signed object or produce a new, clearly attributed statement; it should not pretend signatures survive arbitrary translation.

Sep 22, 2026

Note:

I don't understand this.
Is this criticizing the current
note format? Can it be more
explicit?

Sep 22, 2026

Sep 22, 2026

Note:

agreed. forget about SLSA.
I didn't realize it was
referenced in the valley.

Sep 22, 2026

Sep 22, 2026

Note:

I worry that the-valley playing
with this idea is too prominent.
I would be curious to explore other
creative ways to explore increasing
throughput, including having no checks
on main. Or having no main!!

Sep 22, 2026

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Hypothesis: a hybrid may improve some mixed workloads. Reuse unchanged per-check evidence first. Speculatively execute only invalidated checks on likely future landings. Reduce speculation when failures or conflicts rise. Reserve an integration opportunity for changes that repeatedly lose races. The hybrid adds scheduling and bookkeeping costs, and failed speculation can waste execution. It should be compared experimentally; there is no basis yet for claiming one queue policy works best for every repository.

Oct 3, 2026

Note:

I'd like to hold on to this
thought

Oct 3, 2026

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test observations too, without claiming they are hardware attestations. The current verdict [R014] rejects observations whose calculated age reaches the validity window. Static inspection shows no rejection of negative age. A future timestamp is consequently a concrete test case. The larger issue is defining acceptable clock skew, which clock supplies time, and whether a trusted receipt or challenge is required. A sender's signature authenticates its stated time; it does not make the clock accurate. Not every nondeterministic check belongs in the same category as a live API test. Luo and colleagues studied flaky-test fixes and identified multiple causes of unstable outcomes. A race or test-order dependence can exist inside an otherwise isolated environment. Marking the wrapper nix does not eliminate that behavior. V26 — Luo et al., 2014

Oct 3, 2026

Note:

I'm generally not in love with this time based thing. I guess if we want to have p it we could use a TSA or the thing w/ witnessing.

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an equality-based result. A statistical benchmark may need a sample size, distribution summary, and acceptance rule. A live-service observation needs environment and time context. A flaky deterministic check should trigger investigation of its purity claim, rather than acquire an arbitrary expiration that conceals the defect.

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Note:

Strongly agree.
I'm not sure live services are really in scope here.

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There is a related crash boundary in the existing integrator. land [R031] moves the protected ref before storing the transfer evidence. It reports subsequent failures, but a crash can still leave a landed change without the intended durable record. Composing and signing before the ref move reduces one failure class; it does not atomically publish evidence alongside the landing.

Oct 3, 2026

Note:

File bug?

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Recommendation: investigate a Git ref transaction covering the target and evidence refs, or a durable intent record with recovery. For an external log, specify a two-stage protocol with a recoverable intermediate state. The experiment is a process kill at every persistence boundary, followed by a new controller reconstructing what happened. "Every landing has evidence" should be tested as a storage invariant.

Oct 3, 2026

Note:

I think we need to look at the valley holistically and document the things we don't want to allow actors to hide.

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Finally, sampling without a response mechanism only produces observations. Useful operation requires a disagreement state, preservation of both runs, diagnosis of nondeterminism, and a rule for pausing dependent decisions. Automatic punishment on first mismatch risks treating an honest witness that exposed a flaky check as the attacker.

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Note:

I want to emphasize that perfect security is not the goal nor do I think the current set of design choices is intended for. Much of the initial choices of this playful ~~experimental~~ platform is determinism and observability and immutability ^{so} that we can answer all empirical questions.

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controllers, and a governed identity registry. These choices already produce useful machinery, but they do not yet form the complete system described by the later

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Note:

Git is just the normal thing
but I don't want tight coupling
to it. I suspect there's 1) the
versioning persistence substrate and 2)
the change management vx(model,
where for instance 1) seems for
superior.

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integrator publishes outcomes. Larger claim still requiring work: Every historically important transition can be recovered after bus loss.

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Note:

This was an agentic suggestion that I did not attend to but which I'm not sure of as a requirement for the eventing architecture. For instance I'm intrigued to merged eventing/datasore models like firebase.

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Larger claim still requiring work: A landing and its durable decision record survive every interruption as one complete result.

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Note:

I suspect what's lost in the understanding of the spirit of this approach is that it accepts potential false negatives to accelerate throughput on the hypothesis that subsequent fixes are cheap.

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An event-sourced system retains the events needed to reconstruct application state. Fowler's account also identifies two replay hazards: repeating external updates and substituting present-day external query results for historical ones. Event sourcing therefore requires a replay policy for effects and observations, not just a stored sequence. D02: Event Sourcing Git can carry an event store, but its ordinary commit graph does not automatically record every action taken against a repository. A commit says what tree it names and which commits precede it. A branch ref says which object it currently names. Those are different facts from "this ref moved from A to B in this accepted push." The same final refs can result from several different sequences of pushes.

Oct 3, 2026

Note:

I think about this quite a lot at the moment. What is the right model to maximize throughput and enable collaboration of multiple actors on a project? GDoc, git, DD, are all examples of this.

I suspect that the current SPLC is highly inefficient in the agentic era.

Oct 3, 2026

The existing ref replay primarily serves the first promise. The S4 "why did X occur" requirement needs the second. Reliable deployment and notification need the third. None follows from the others.

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Note:

This looks like an argument
→ centralize mutation operations through
a log.

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Git offers transactions over multiple ref updates through `update-ref --stdin`. Its documentation also warns that a concurrent reader may observe only some modifications even when the transaction's updates were accepted together. That makes multi-ref transactions a candidate tool, with reader and crash behavior to test against the selected ref backend. They are not a blanket claim of database-style snapshot isolation. D04: git-update-ref

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Note:

I know there's a lot of risk
to it, but I wonder if the
volley should just expose its
own change API; and hide
the backing repo for updates.

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or Temporal.

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Note:

I'm not familiar with temporal or how it's relevant.

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The durability requirement [R043] is unusually concrete: integrated work should reach independent copies, including an offsite copy, within minutes, and a copy counts only after a verified restore. That is a stronger starting point than merely configuring backup software. The remaining ambiguity is what must be restored together.

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Recommendation. Name three observable states for an accepted change: locally committed, independently replicated, and included in a verified recoverable snapshot. This need not put a remote round trip on the integration path. It makes the allowed exposure window visible. The requirement's phrase "integrated means replicated ... within minutes" otherwise mixes an immediate status with an eventual obligation.

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Note:

This needs follow up discussion.
Ideally the-volley would just
have highly available/durable
implementations.

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Opinion. Keep persistent principals few, but give every run and every privileged effect a distinct durable identity. Keep the signer, the claimed producer, the delegating authority, and the verifier as separate fields. Collapsing them makes records simpler to print and harder to trust.

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Note:

I think this is fine. I also
think that a trajectory's initial
context and permissions are reusable
and might need a name.

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4.9. Federation should exchange evidence while preserving local authority

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Note:

As mentioned earlier. An idea I like is a local first API surface. like Firebase but where the mutation path and the read path are separate schemas. The mutation schema is a fixed set of mutation operations that can stack.

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5. Knowledge, agents, and outcomes

Hypothesis: small project-specific verbs such as "show current decision," "explain blockers," and "show evidence for completion" will provide more reliable value initially than an elaborate autonomous memory manager. Test that hypothesis under equal tool-call and context budgets.

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Note:

I think comprehensive documentation is very hard but made much easier thanks to agents and genesis, design, and user documentation serves a purpose for aligning agents and humans.

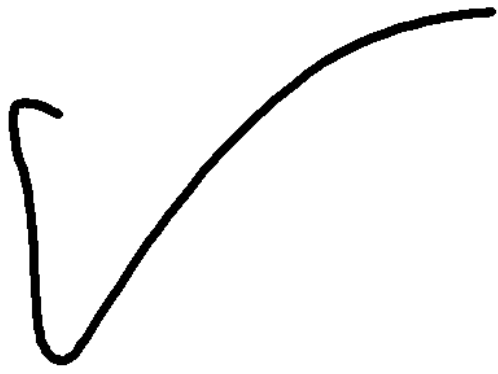
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scheduling. A08: Erol et al., 1994 Recommendation: initially let agents propose decompositions while a deterministic scheduler executes an accepted graph. Each proposed decomposition should state its parent condition, its assumptions, why the children are sufficient, and what would invalidate the plan. A new child should consume a share of the parent's budget; adding an edge should never mint new spending authority. This is a recommendation for bounded automation, not a demand for approval of every low-impact task. Pin the accepted objective and acceptance procedure, including which authority may revise them. An agent's proposed amendment must remain distinguishable from the rule currently deciding success.

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Note:



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Bainbridge's Ironies of Automation explains a central difficulty of supervision: automation may leave humans responsible for exceptional intervention while depriving them of the practice and ongoing involvement needed to intervene well. It is a conceptual analysis of industrial automation, not a quantified study of coding-agent review. Its relevance is the failure mode it makes visible. A25: Bainbridge, 1983

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Note:

I love this observation. It's a central tension that I agree with and ignoring it could be fatal over time.

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Assessment: the-valley should learn experimental discipline from these sources without choosing one headline score as its objective. The system exists to produce outcomes while preserving control and understanding. Patch success, operator time, latency, expenditure, and later rework are separate measurements. A theory-reconstruction experiment

Oct 3, 2026

Note:

I agree the valley should measure and evaluate trajectory success and also continuously update itself on model performance data from outside sources

Oct 3, 2026

6. Synthesis and a research agenda

Ordinary systems are stronger baselines than the premise suggests. Concurrent merge validation, portable execution, distributed collaboration records, and policy stored in a repository all have working precedents. the-valley should be compared with configurations that use these mechanisms well. That makes a positive result more meaningful and can identify components worth adopting. Chapter 2 supplies those baselines.

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Note:

I think there's a strong NIH bias in the valley based on the lower cost of materializing and changing our own code in this agentic moment. If we can accelerate development we have agency to leverage that on the whole stack.

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A malicious local attester must reveal a modified tool hash More defensible statement: A signer can claim an approved hash without executing that tool Evidence and next check: Protocol counterexample in chapter 3; test a directly fabricated signed statement

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Note:

I think scaling up to paranoid organizational sec risks the attestation model and nix env lay the ground work for trusted builds in TEEs.

Oct 3, 2026

6.5 Experiments that would change the design

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Note:

I really like these as success metrics for the -Valley. Maybe the hierarchy of knowledge graph goes something like

- Purpose: users, usecases, goals/mission
- Requirements: high throughput, valuable outcomes, efficient use of attention

Success metrics: things to kill drive/validate

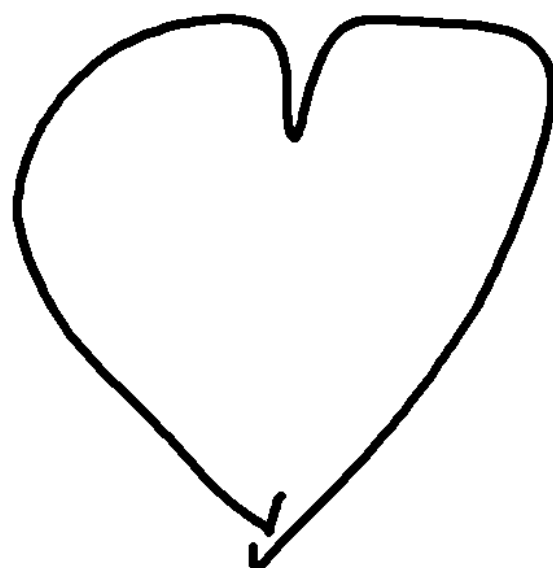
architecture: Design, components, interfaces

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6.6 Measurement discipline

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Note:



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7. Reading guide and glossary The most useful way to study this material is to move between a primary source, a concrete failure case, and the corresponding part of the valley. The sequence below is designed to develop judgment about mechanisms rather than familiarity with names. The source registers provide direct links and state which portions were inspected for this report.

Oct 3, 2026

Note:

I'd love a careful agent's
tutoring approach to approaching
this content.

Oct 3, 2026
